Total No. of Questions : 8]

SEAT No. :

PB-2507

[Total No. of Pages : 2

[6263]-393

B.E. (Artificial Intelligence and Data Science)

BIG DATA ANALYTICS

(2019 Pattern) (Semester - VIII) (417532B) (Elective - V)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Use of non programmable electronic calculator is allowed.***
5) ***Assume suitable/standard data if necessary.***

Q1) a) What are the primary methods and functions for importing and exporting data in R and how can they be utilized effectively? **[8]**

b) How can data analysts detect and address dirty data using visualizations and statistical techniques in R and what are the implications of anomalies in datasets on decision-making processes? **[10]**

OR

Q2) a) With the help of neat diagram explain the phases of Data Analytics life cycle. **[8]**

b) What are the different attribute types in data analysis, and how are they categorized? How does R handle various data types, such as numeric, character, logical, and factors? **[10]**

Q3) a) Explain the following terms with respect to Exploratory data analysis. **[10]**

i) Data Sourcing
ii) Data Cleaning
iii) Univariate analysis
iv) Bi-variate/Multivariate analysis

b) What are the essential steps in data exploration and how do they contribute to uncovering insights and patterns within a dataset? **[7]**

OR

Q4) a) How do ensemble methods such as bagging, boosting, AdaBoost and Random Forest contribute to improving classification accuracy in machine learning models? **[10]**

b) How does the utilization of a confusion matrix aid in the evaluation and selection of models. **[7]**

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Q5) a) What are the challenges associated with visualizing big data and how do these challenges impact the effectiveness and efficiency of data analysis and decision-making processes? **[10]**

b) How does Tableau facilitate effective data visualization and what are some advanced techniques or features within Tableau that can be utilized to create insightful and interactive visualizations for complex datasets? **[8]**

OR

Q6) a) What are the key features and functionalities of the Google Chart API and how does it enable developers to create dynamic and interactive charts and visualizations for web applications? **[8]**

b) What are the various types of data visualization techniques available to data analysts? **[10]**

Q7) a) In what ways does financial data analytics leverage big data technologies to drive insights, mitigate risks and enhance decision-making processes within the financial industry? **[10]**

b) How does Apache HBase contribute to efficient data storage and retrieval in big data environments **[7]**

OR

Q8) a) How does the Semantria tool streamline the data collection process and what features or capabilities does it offer that make it a valuable asset for businesses seeking to gather and analyze large volumes of unstructured data from various sources? **[10]**

b) How does Mozenda serve as an effective data filtering and extraction tool? **[7]**

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